

DOMAIN FORENSIC ANALYZER - PROJEKTFORTSCHRITT OVERVIEW

AKTUELLER STATUS (Stand: Phase 2.4 Abgeschlossen)

IMPLEMENTIERTE CORE-MODULE

1. DNS Foundation (dns_analyzer.py)

- **Status:** Production-Ready
- **Features:** IPv4/IPv6 Resolution, MX/NS Records, Reverse DNS
- **Performance:** ~4-5s execution time
- **Output:** Strukturierte DNS-Intelligence

2. CDN Detection (cdn_detector.py)

- **Status:** Production-Ready
- **Features:** Provider Detection (Cloudflare, GitHub, etc.), Geolocation, ASN-Info
- **Performance:** ~0.1-0.2s execution time
- **Output:** Infrastructure Classification

3. Network Intelligence (network_intelligence.py)

- **Status:** Production-Ready
- **Features:** Traceroute Analysis, OPSEC Assessment, Connectivity Testing
- **Performance:** ~30-45s execution time
- **Output:** Network Path Analysis, Route Intelligence

4. Asset Discovery (subdomain_scanner.py)

- **Status:** Production-Ready
- **Features:** Multi-threaded Subdomain Enumeration, Risk Categorization
- **Performance:** ~0.3-0.4s execution time
- **Output:** Sensitive Asset Detection (admin, api, dev interfaces)

5. Threat Intelligence (securitytrails_client.py)

- **Status:** Demo-Mode (API-Key Required)
- **Features:** Historical Subdomain Data, Domain Intelligence
- **Performance:** Instant (demo), ~30s (live)
- **Output:** Intelligence Summary, Risk Scoring

ZUSÄTZLICHE INFRASTRUKTUR

Result Aggregator (result_aggregator.py)

- **Status:** Production-Ready
- **Features:** UnifiedResult Structure, JSON Export, Risk Scoring (0.0-10.0)
- **Output:** Standardized Professional Reports

Enhanced Error Handling

- **Timeout Management:** Module-specific timeouts (30-180s)
- **Threading Integration:** Professional error recovery
- **Performance Monitoring:** Execution time tracking
- **Graceful Degradation:** Fallback results on failure

Professional Logging

- **Loguru Integration:** Structured logging with rotation
- **Performance Metrics:** Detailed execution tracking
- **Debug Information:** Asset extraction details
- **Portfolio-Ready:** Industry-standard logging practices

ENTDECKTE ABER NICHT AKTIVIERTE MODULE

certificate_analyzer.py ?

- **Status:** Existiert, aber nicht in Execution-Chain
- **Potential:** Certificate Transparency, SSL Analysis
- **Integration:** Muss analysiert und aktiviert werden

integration_test.py ?

- **Status:** Existiert
- **Purpose:** Vermutlich API-Testing Framework
- **Action:** Analyse erforderlich

AKTUELLE FUNKTIONALITÄT

COMPLETE ANALYSIS WORKFLOW

Input: domain.com

- DNS Foundation Analysis (~5s)
- Infrastructure Classification (~0.2s)
- Network Intelligence (~35s)
- Asset Discovery (~0.3s)
- Threat Intelligence (~0s demo)

Output: Professional Report + JSON Export (~40s total)

SUCCESSFUL TEST DOMAINS

-  **github.com**: 16 assets, 4 sensitive, LOW risk
-  **stackoverflow.com**: 6 assets, 1 sensitive, MINIMAL risk
-  **Error Handling**: Graceful degradation validated

PROFESSIONAL OUTPUT FEATURES

- **Clean Status Messages**: "COMPLETE", "FAILED", "TIMEOUT"
- **Performance Tracking**: Per-module execution times
- **Risk Assessment**: Automated scoring with recommendations
- **Portfolio-Ready**: No emojis, professional formatting

PHASE 2.5 - NÄCHSTE PRIORITÄTEN

IMMEDIATE: API-KEY CONFIGURATION

1. SecurityTrails Live-Integration

- **Priority**: HIGH (already integrated, needs API key)
- **Effort**: 30 minutes
- **Impact**: Historical subdomain intelligence

2. Additional APIs Integration

- **VirusTotal**: Domain reputation, malware detection
- **Shodan**: Network device discovery
- **Censys**: Certificate transparency analysis
- **Effort**: 2-3 hours per API

3. Certificate Analyzer Activation

- **Analyze existing code**: Understand current implementation
- **Integration**: Add to execution chain
- **Testing**: Validate functionality

FUTURE: DNS-FORENSIK-MODUL

NEUES MODUL: CDN-WORKAROUND & HISTORICAL DNS

Trigger-Condition:

```
if cdn_detected and provider_type in ['cdn', 'ddos_protection']:
    execute_dns_forensics_module()
```

FORENSIC METHODS FRAMEWORK:

1. Historical DNS Timeline Analysis

- SecurityTrails historical data
- Passive DNS repositories
- DNS change pattern detection

2. Origin Server Discovery

- Certificate-based discovery
- Subdomain analysis for origin leaks
- HTTP header analysis

3. Infrastructure Evolution Tracking

- CDN adoption timestamps
- Server migration patterns
- Hosting provider changes

OUTPUT FORMAT:

```
{
  "dns_forensics": {
    "cdn_workaround_attempted": true,
    "historical_timeline": [
      {
        "date": "2019-03-15",
        "ip": "95.123.45.67",
        "provider": "Kleinfirma GmbH Stuttgart",
        "confidence": "high"
      }
    ],
    "origin_server_candidates": [...],
    "investigation_leads": [...]
  }
}
```



DEVELOPMENT ROADMAP

PHASE 2.5: API INTEGRATION (1-2 Tage)

1. API-Keys konfigurieren
2. SecurityTrails live testen
3. Additional APIs hinzufügen
4. Integration testing

PHASE 2.6: DNS-FORENSIK-MODUL (3-4 Tage)

1. CDN-Detection-Enhancement
2. Historical DNS Analysis Framework

3.  Origin Server Discovery Methods
4.  Forensic Timeline Generation
5.  Professional Investigation Reporting

PHASE 3: PORTFOLIO FINALIZATION (1 Tag)

1.  Documentation completion
2.  GitHub repository optimization
3.  LinkedIn showcase preparation
4.  Demo scenarios dokumentation

PORTFOLIO VALUE

DEMONSTRATED SKILLS

- **Multi-API Integration:** Professional service orchestration
- **Error Handling:** Industry-standard resilience patterns
- **Security Analysis:** OSINT methodologies, risk assessment
- **Performance Optimization:** Threading, timeouts, monitoring
- **Code Architecture:** Modular, extensible, maintainable design
- **Professional Tooling:** Logging, testing, documentation

READY FOR SHOWCASE

-  **Professional Code Quality:** Production-ready standards
-  **Real-world Applicability:** Actual security use cases
-  **Technical Sophistication:** Advanced programming patterns
-  **Portfolio Presentation:** Clean, documented, demonstrable